

Alexander Tiu

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EDUCATION: Southern Methodist University	Dallas, TX
Bobby B. Lyle School of Engineering	Expected Graduation: 2028
Bachelor of Science in Electrical and Computer Engineering	GPA: 3.98

RELEVANT COURSES: Linear Algebra, Differential Equations, Signals & Systems, Embedded Systems, Programming in C++, Circuit Analysis, Digital Computer Logic Design

AWARDS:

President's Scholar – SMU's highest merit scholarship recognizing exceptional academic and leadership excellence
Nunn-Perry Award Nominee (2026) – Lead Undergraduate Researcher, SMU Research Team
University Honors Program | Lyle Discovery Scholarship

PERSONAL PROJECTS:

Wi-Fi Controlled RC Car | ESP32, Embedded C++, Motor Drivers May 2025– Present

- Built a Wi-Fi operated RC car using an ESP32 microcontroller with browser-based remote controls.
- Implemented bidirectional motor control logic using GPIO signals and H-bridge motor drivers.
- Integrated embedded software and hardware debugging to support real-time wireless operation

RESEARCH EXPERIENCE:

Quantum Computing Researcher, Darwin Deason Institute of Cybersecurity May 2026 - Present

- Develop theoretical models of frequency-bin photonic quantum systems using MATLAB, linear algebra, and quantum optics.
- Simulate Mach–Zehnder modulator (MZM) architectures to investigate frequency-bin qudit manipulation, photonic quantum gates, and interference phenomena.
- Analyze unitary transformations, interference effects, and post-selection techniques for integrated quantum photonic devices.
- Collaborate with graduate researchers to explore novel quantum information processing architectures and theoretical device performance.

Machine Learning Researcher & Project Lead, Darwin Deason Institute of Cybersecurity January 2025 – May 2026

- Conducted graduate-level research on AI-driven threat detection and simulator-based analytical systems.
- Developed explainable machine learning models using XGBoost, SHAP, and large language models on multi-million-record maritime datasets.
- Led development of an LLM-based explainability pipeline, designing project architecture and evaluation methodologies for operational decision support.
- Presented research at the Naval Applications of Machine Learning (NAML) Conference (2026); project nominated for the Nunn-Perry Award.

Mustang Rocketry August 2025 – Present

- Designing and constructing an L1-class high-power rocket for certification under Tripoli Rocketry Association standards.
- Collaborate in a team-based engineering environment for high-power rocket development

Lyle School of Engineering Student Council, Representative August 2025 - Present

- Represent student body within the Lyle School of Engineering, providing input to faculty leadership on curriculum and program decisions.

SKILLS:

- Programming Languages: Python, MATLAB, C++
- Machine Learning: XGBoost, scikit-learn, SHAP, model interpretability, feature engineering
- PCB Design: Altium Designer (Certified)
- Tools: Github, Visual Studio Code
- SolidWorks (Coursework in Progress)